MILITARY SPECIFICATION

Computer Set, Navigational

AN/ASN-42(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall, when integrated with a doppler ground velocity sensor, provide accurate navigational outputs for Naval aircraft.

1.2 Classification - The Navigational Computer Set AN/ASN-42 shall consist of the following items:

DESCRIPTION	TYPE DESIGNATION	APPLICABLE PARAGRAPH
Gyroscope Assembly	CN-793/ASN-42	3.4.1
Mounting,Gyroscope Assembly	MT-2200/ASN-28	3.4.2
Computer, Navigational	CP-632/ASN-42	3.4.3
Mounting	MT-2959/ASN-42	3.4.4
Power Supply	PP-3100/ASN-42	3.4.5
Mounting	MT-2962/ASN-42	3.4.6
Control, Navigational Computer Set (Alignment Control)	C-3833/ASN-42	3.4.7
Converter, Signal Data	CV-1227/ASN-42	3.4.8
Mounting	MT-2961/ASN-42	3.4.9
Amplifier, Electronic Control	AM-3113/ASN-42	3.4.10
Mounting	MT-2960/ASN-42	3.4.11
Control, Gyroscope Assembly	C-3832/ASN-42	3.4.12
Control, Inertial Navigation	C-4061/ASN-42	3.4.13

1.3 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

Navigator Computer, Doppler/Air Mass, AN/ASA-47
Attitude & Heading Reference System (AHRS), AN/ASN-37
Autopilot (PB-20 Mod)
Indicator, BDHI, 4 each, ID-663/U or equal
Indicator, MM-4
Maneuver Monitor, MX-2230 ( )/ASQ
AN/APS-80 Radar, 2 eacn

FSC 5826

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1.3 Associated Equipment (Continued)

Remote Compass Transmitter, Type ML-1
AN/ARN-21A or AN/ARN-52 TACAN
MW-2 (MS 28094-1) Compass Repeater Indicator
Indicator Group AN/ASA-16
Navigational Plotter, PT-396/AS
Flight Station Plotter, PT-434/ASA-13
Power Supply and Relay Unit
Doppler Radar Set AN/APN-122(V)
Doppler Radar Set AN/APN-153(V)

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-E-7894	Electric Power, Aircraft; Characteristics of
MIL-E-17555	Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment; General Specification for

STANDARDS

Federal

FED-STD-595, March 1, 1956	Colors

Military -

MS-28073	Meter, Time Totalizing, 115 Volt 400 Cy Rev B

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2.2 Availability of Documents

(1) When requesting specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, excluding cables shall be a minimum consistent with good design and shall not exceed 180 pounds.

3.2.2 Reliability

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 50 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operating Life - The equipment shall have a minimum total operating life of 3000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.2.3 Reliability In Mean Time Between Failures (MTBF) - For the test program under 4.4.3, the mean (operating) time between failures shall be 80 hours. Procedure II of Specification MIL-R-23094 shall be used.

3.2.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meters in accordance with Specification MIL-M-7793. Elapsed time indicators shall be provided for both the Doppler System and the Navigational Computer Set. Both indicators shall be located in the Signal Data Converter and shall have a maximum scale of 10,000 hours.

3.2.3 Cabling and Connections -

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

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3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing. The color of the control panels shall conform to the following:

Console Control	Color of Lamp Holder	Plastic Outer Coating Color	Backing Plate Color	Knob Color	Lamp Holder Filter Color
C-3833/ASN-42	Gray *	*Gray	*Gray	Black	Clear
C-3832/ASN-42	Gray *	*Gray	*Gray	Black	Clear
C-4061/ASN-42	Black	Black	Black	Black	Aviation Red

* Color number 36231 Federal Standard Number 595 (Lusterless Dark Gull Grey)

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° C ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power	115 ±1.0 V AC and/or 27.5 ±0.5 V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein:

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3.2.10.1 Vibration - When normally mounted (with vibration isolators in place, if any), the equipment shall operate satisfactorily when subjected to continuous vibration within the frequency range of 5 to 500 cps, and having a double amplitude of 0.036 inch in any direction, with a maximum acceleration of 10g (except that the acceleration may be reduced to 2g for the Gyroscope Assembly), whichever is the limiting value. Except for the Gyroscope Assembly, equipment which is normally mounted on vibration isolators shall operate satisfactorily with isolators removed when subjected to 2gs, .01 inch double amplitude vibration within the frequency range of 5 to 500 cps in accordance with curve II of MIL-E-5400.

3.2.10.2 Temperature - Five (5) pounds per minute of cooling air at not more than 110° must be available to Gyroscope Assembly, CN-793/ASN-42 for any operation in ambient of over 110°F. Cooling air is not supplied by the Inertial Navigation equipment.

3.2.10.3 Shock - This equipment shall operate satisfactorily when subjected to the shock conditions specified in MIL-E-5400, except that the shock amplitude of 15g will be reduced to 10g for the Gyroscope assembly. In addition, the equipment and its mounting shall be so designed and constructed that the equipment shall not tear loose from its mounting when subjected to an anticipated ditching shock of 16g along the longitudinal axis of the airplane.

3.2.11 Warm-up and Align Time - The equipment shall be capable of operating within the specified performance tolerances after an align time of 20 minutes at temperatures above 25°C, 30 minutes for temperatures from 0°C to and including 25°C; 45 minutes for temperatures from -45°C to and including 0°C. The above requirements apply for initial alignment and operation within the latitude of North 70° and South 70° and after initial rough alignment to within ±5° of local level and true North.

3.2.12 Input Power Requirements - This equipment shall meet all applicable requirements of Specification MIL-E-7894 except as modified herein.

3.2.12.1 AC Voltage Limits - The line to neutral steady state voltage will be nominally 115 volts plus or minus 5 volts with fault limits of 103 to 129 volts. These faults may exist for an indefinite length of time during which the system may operate with decreased performance, but shall not be damaged.

3.2.12.1.1 Voltage Transient Limits - Voltage transients will be within the transient envelope shown on Figure 1. Transients which exceed the fault limits of Paragraph 3.2.12.1 will exist for a maximum of 5 volt-seconds. The equipment shall not be damaged when these transients occur.

3.2.12.1.2 Overvoltage - Overvoltage will be limited in accordance with the overvoltage curve shown in Figure 1. Overvoltage conditions exceeding the fault limits of Paragraph 3.2.12.1 will exist for a maximum of 5 volt-seconds.

3.2.12.1.3 Phase Balance - Voltage unbalance will not exceed 4% under normal conditions. All three-phase equipment shall present a balanced load insofar as practicable.

3.2.12.1.4 Voltage Modulation - Voltage modulation will not exceed 2%.

3.2.12.1.5 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified.

3.2.12.2 AC Frequency Limits - The normal power system frequency will be 380 to 420 CPS. The over/under frequency limits are 360 ±5 and 425 ±5 CPS. Under certain ground operating conditions, these frequency limits may exist for 5 seconds maximum and under these conditions, no damage to the equipment shall occur.

3.2.12.2.1 Harmonic Content - No harmonic will exceed 5%. The crest factor will be 1.41 ±10%.

3.2.12.2.2 Frequency Modulation - The contractor shall declare any susceptibility of the equipment to frequency modulation and shall indicate the tolerable amplitude limits over any critical band.

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3.2.12.2.3 Underfrequency Protection - The equipment shall not be damaged by frequency voltage value combinations falling within the shaded envelope defined by Figure 3.

3.2.12.3 DC Voltage Limits - The steady state voltage limits for DC power will be within the limits of 24 to 30 volts, except that the limits for the "Emergency (generator out)" condition will be between 14 and 29 volts.

3.2.12.3.1 Transient Limits - Voltage transient limits will be within the envelope shown on Figure 2. The equipment shall not require recycling when these transients occur.

3.2.12.3.2 Overvoltage - The equipment shall not be damaged when subjected to voltages above the normal and below the overvoltage limit of the curves shown on Figure 2.

3.2.12.4 Power Interruptions - During and at the beginning of operation AC power may be interrupted in a feeder to one or more phases. The equipment shall not be damaged by such failures and shall stay within specified accuracy following the occurrence of two .060 second interruptions per flight; interruptions to be time separated by at least five (5) minutes. Decreased accuracy is allowable if such failures are more frequent. Any transient beyond lower transient envelope of Figure 1 shall be considered a power interruption.

3.2.12.5 Total Power - The total AC primary input power requirement for this equipment shall not exceed 1.4 KVA.

3.2.13 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.2.14 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments, shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be approved by the procuring agency.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 General - The Navigational Computer Set shall be operated in the aircraft with a Doppler Radar System (AN/APN-122(V) or AN/APN-153(v)) to provide an integrated Inertial Doppler Navigation System. The Navigational Computer Set shall operate in one navigation mode and several attitude and heading reference modes. Navigational information shall include present position in terms of latitude and longitude, north-south and east-west miles travelled, aircraft velocity, true and magnetic heading, magnetic variation, and aircraft pitch and roll. The various principal modes of operation shall be:

(1) Inertial Doppler
(2) Slave
(3) Free
(4) Compass

3.3.1.1 Inertial Doppler Mode - In this mode, the Navigational Computer Set shall utilize inputs from the Doppler system to give navigational information and attitude and heading information. In the event that the Doppler system goes into the memory mode or the Doppler Decoupling signal is received from the Power Supply and Relay unit, the Navigational Computer Set shall continue to supply navigational information. In the Inertial Doppler mode the Navigational Computer Set shall provide the following outputs:

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(1) Doppler Heading Velocity
(2) Doppler Drift Velocity
(3) Present Latitude
(4) Present Longitude
(5) North-South miles travelled
(6) East-West miles travelled
(7) Inertially derived North-South velocities
(8) Inertially derived East-West velocities
(9) Aircraft Roll Angle
(10) Aircraft Pitch Angle
(11) Aircraft True Heading
(12) Aircraft Stabilized Magnetic Heading
(13) Aircraft Clutched Heading
(14) Magnetic Variation
(15) Doppler Memory Signal
(16) Navigational Computer Set Failure Indications
(17) I-D Mode Signal
(18) Rate of Turn Signal
(19) Magnitude of the difference between Doppler N-S and Inertial N-S velocity components.
(20) Magnitude of the difference between Doppler E-W and Inertial E-W velocity components.

3.3.1.2 Slave Mode - In the Slave Mode the Navigational Computer Set operates as a pendulous vertical and provides magnetic heading and aircraft roll and pitch. Velocity and acceleration corrections are applied to assist in maintaining vertical when velocity information is available. Earth's rate corrections are always applied to assist in maintaining vertical. True Heading can be obtained if magnetic variation is set into the system. The outputs in the Slave Mode are as follows:

(1) Aircraft Roll
(2) Aircraft Pitch
(3) Aircraft Stabilized Magnetic Heading
* (4) Aircraft True Heading
** (5) Doppler Heading Velocity
** (6) Doppler Drift Velocity
(7) Aircraft Clutched Magnetic Heading
* (8) Magnetic Variation, Hand Set
(9) Doppler Memory Signal
(10) Navigational Computer Set failure indications
(11) Rate of Turn Signal

* Correct only if Magnetic Variation is properly set.
** Available only if Doppler System is turned on.

3.3.1.3 Free Mode - In the Free Mode the Navigational Computer Set operates as a pendulous vertical and provides selected grid heading and aircraft roll and pitch. Velocity corrections and earth's rate corrections will not be used to assist in maintaining vertical; however, earth's rate corrections will be applied to the heading to correct for earth's rotation. Rate of turn sensing will be used to minimize false erection during turns. The outputs in the Free Mode are as follows:

(1) Aircraft Roll
(2) Aircraft Pitch
(3) Aircraft selected Grid Heading (appears on Mag. Heading and True Heading outputs)
** (4) Doppler Heading Velocity
** (5) Doppler Drift Velocity
(6) Aircraft clutched grid heading (on clutched Mag. Heading output)
*(7) Magnetic Variation, Hand Set
(8) Doppler Memory Signal
(9) Navigational Computer Set failure indications

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(10) Rate of Turn Signal
(11) Free Mode Signal

* Correct only if magnetic variation is properly set.
** Available only if the Doppler System is turned on.

3.3.1.4 Compass Mode - In the Compass Mode the Navigational Computer Set operates primarily to repeat raw flux valve heading, even in the presence of a Navigation Computer Set failure indication. If this mode is selected when the Navigational Computer Set is functioning normally, the operation will be the same as in the Fast Erect Mode except that magnetic heading is repeated flux valve heading. In this case the true heading output is not driven and a Navigational Computer Set failure indication will be given. The outputs in the Compass Mode are as follows:

(1) Raw Flux Valve Magnetic Heading (appears on Mag. Heading output)
*** (2) Aircraft Roll
*** (3) Aircraft Pitch
(4) Aircraft Clutched Raw Flux Valve, Magnetic Heading
** (5) Doppler Heading Velocity
** (6) Doppler Drift Velocity
* (7) Magnetic Variation, Hand Set
(8) Doppler Memory Signal
(9) Navigational Computer Set failure Indication

* Correct only if magnetic variation is properly set
** Available only if the Doppler System is turned on
*** Available only if there is no Navigational Computer Set

3.3.2 System Accuracies - The equipment shall be capable of the following:

3.3.2.1 Miles Traveled North/South and East/West - Miles travelled North/South and East/West shall be calculated. The Inertial-Doppler Mode navigational error shall not exceed 2.0 nautical miles CEP per hour of operation following normal ground alignment between the latitudes of N70° to S70°. In the Inertial Mode the navigational error shall not exceed 2.0 nautical miles CEP per hour for the first four hours of operation between the latitudes of N70° and S70°.

3.3.2.2 True Heading - Normal Mode - True heading output shall be supplied from the platform synchro with RMS error of (0.15°) + (.04°/HR). True heading outputs from the follow-up servos shall be supplied with an RMS error of (0.25°) + (.04°/HR).

3.3.2.3 Magnetic Heading - Stabilized magnetic heading shall be supplied as an output with a maximum error of ±0.25° after compensation, and excluding any remote compass transmitter errors. Compensation shall be accomplished at 24 points.

3.3.2.4 Pitch Angle - Supply a pitch angle output with a maximum error of ±7 min. over a range of ±15° centered about zero degrees and ±12 min. over the range of ±(15° to 45°).

3.3.2.5 Roll Angle - Supply a roll output witn a maximum error of ±7 min. over a range of ±45° centered about zero degrees and ±12 min. over the range of ±(45° to 75°).

3.3.2.6 Clutched Magnetic Heading - Supply clutched magnetic heading outputs with a maximum error of ±0.7 degrees.

3.3.2.7 Magnetic Variation - Calculate magnetic variation to an accuracy of 1/2 degree + .04°/hr RMS excluding any remote compass transmitter errors. This is computed in the I-D only.

3.3.2.8 N-S-E-W Velocities - 1 knot or 1/2 percent, whichever is greater.

3.3.2.9 Rate of Turn Signal - Shall be given when rate of turn is 18 degrees/min., ±3 degrees/min.

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3.3.2.10 Latitude and Longitude - The latitude and longitude as computed by the Inertial-Navigation System shall not be in error by more than 2 nautical miles per hour, CEP as specified in paragraph 3.3.2.1.

3.4 Detail Requirements - This unit shall meet the following requirements:

3.4.1 Gyroscope Assembly CN-793/ASN-42 -

3.4.1.1 Function - The Gyroscope Assembly establishes the stable reference, with respect to the earth, for the measurements of acceleration and aircraft attitude and heading.

3.4.1.2 Form Factor - The maximum allowable dimensions of the Gyroscope Assembly shall be 12.75 inches high by 14.50 inches wide by 16.56 inches long.

3.4.1.3 Weight - The weight of the gyroscope assembly shall not exceed 35.0 pounds.

3.4.2 Mounting, Gyroscope Assembly MT-2200/ASN-28 - This unit shall meet the following requirements.

3.4.2.1 Function - The gyroscope assembly mount isolates the gyroscope assembly from aircraft vibration in a manner such that the gyroscope assembly angular motions are minimized.

3.4.2.2 Form Factor - The maximum allowable dimensions of the gyroscope assembly mount are 8.75 inches high, 12.75 inches wide and 16.56 inches long.

3.4.2.3 Weight - The weight of the gyroscope assembly mount shall not exceed 8.0 pounds.

3.4.3 Computer, Navigational CP-632/ASN-42 - This unit shall meet the following requirements:

3.4.3.1 Function - The primary function of the Navigational Computer is to compute aircraft velocity, present position and distance travelled. In addition, it computes the Coriolis corrections for the measured acceleration and the angular rates required to maintain the stable element's local geographic vertical and true north orientation. The electronic amplifiers and circuitry in the Navigational Computer are also used to accomplish initial levelling and gyrocompassing of the stable element and to maintain a pendulous vertical during the heading modes of operation.

3.4.3.2 Form Factor - The maximum allowable dimensions of the navigation computer are 10.91 inches high, 10.38 inches wide and 14.81 inches long.

3.4.3.3 Weight - The weight of the navigation computer shall not exceed 41.7 pounds.

3.4.3.4 Electrical Connections - The following voltages will be supplied by the power supply and relay unit and shall not be subject to interruptions:

45.0 ±1% Volts DC
20.0 ±1% Volts DC
28.0 ±1% Volts DC
-45.0 ±1% Volts DC

3.4.4 Mounting, Navigational Computer MT-2959/ASN-42 - This unit shall meet the following requirements.

3.4.4.1 Function - The navigational computer mount isolates the navigational computer from aircraft vibrations.

3.4.4.2 Form Factor - The maximum allowable dimensions of the navigational computer mount are 1.5 inch high, 11.59 inches wide and 15.12 inches long.

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3.4.4.3 Weight - The weight of the navigational computer mount shall not exceed 4.3 pounds.

3.4.5 Power Supply PP-3100/ASN-42 - This unit snall meet the following requirements.

3.4.5.1 Function - The function of the Power Supply is to provide the a-c and d-c power required by the Inertial Navigation System.

3.4.5.2 Form Factor - The dimensions of the power supply shall not exceed 5.94 inches high, 4.57 inches wide and 14.00 inches long.

3.4.5.3 Weight - The weight of the power supply shall not exceed 15.6 pounds.

3.4.6 Mounting MT-2962/ASN-42 - This unit shall meet the following requirements:

3.4.6.1 Function - The power supply mount shall isolate the power supply from aircraft vibration.

3.4.6.2 Form Factor - The dimensions of the power supply mount shall not exceed 1.31 inches high, 5.13 inches wide and 14.25 inches long.

3.4.6.3 Weight - The weight of the power supply mount shall not exceed 1.5 pounds.

3.4.7 Control, Navigational Computer Set C-3833/ASN-42 - This unit shall meet the following requirements:

3.4.7.1 Function - The Navigational Computer Set Control (Alignment Control) contains the following controls and displays: The operate controls, magnetic variation control, set magnetic variation display, advisory display, and failure indicator. These controls and displays permit the operator to propersly align and operate the Navigational Computer Set when in the I-D mode and to set magnetic variation wnen in other modes of operation.

3.4.7.2 Form Factor - The dimension of the alignment control shall not exceed 6.00 inches high, 5.75 inches wide and 6.45 inches long.

3.4.7.3 Weight - The weight of the alignment control shall not exceed 5.6 pounds.

3.4.7.4 Controls and Indicators - The Alignment Control shall consist of the following controls and displays.

3.4.7.4.1 The Operate Control - Tne operate control permits selection of align or operate condition.

3.4.7.4.2 Magnetic Variation Control - This control permits hand setting of magnetic variation.

3.4.7.4.3 Set Mag Var Display - This display indicates when magnetic variation must be hand set.

3.4.7.4.4 Magnetic Variation Display - This display permits reading of either computed or hand set variation.

3.4.7.4.5 Advisory Displays - The advisory displays shall consist of a Pilot and Align Indicator and a Ready Meter. The Pilot Indicator snall indicate that the Inertial Navigation System is not in the I-D mode; the Align Indicator shall indicate that the Inertial Navigation System is undergoing alignment; the Ready Meter shall indicate that the system has completed alignment in the I-D mode.

3.4.7.4.6 Failure Indicator - This display indicates malfunction of the Navigational Computer Set.

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3.4.8 Converter, Signal Data CV-1227/ASN-42 - This unit shall meet the following requirements:

3.4.8.1 Function - The Signal Data Converter accomplishes most of the interconnections between the Inertial Doppler Navigation System and other equipment in the aircraft. This unit also contains the various heading servo follow-ups and the circuits for flux valve compensation. The unit also accomplishes the Doppler and Navigational Computer Set interconnection by computing the necessary corrections for the Inertial Doppler Navigation System.

3.4.8.2 Form Factor - The dimensions of the signal data converter shall not exceed 7.32 inches high, 8.59 inches wide and 14.97 inches long.

3.4.8.3 Weight - The weight of the signal data converter shall not exceed 30.5 pounds.

3.4.8.4 Electrical Connections - Connections to external circuits shall be provided as follows:

3.4.8.4.1 Inputs From Associated Equipment -

3.4.8.4.1.1 Magnetic Heading - Magnetic heading shall be supplied from a remote compass transmitter meeting the requirements of MIL-T-19576 and is used in the I-D, slave, and compass modes.

3.4.8.4.1.2 North-South and East-West Velocities - These velocities shall be supplied as 400 cycle AC signals referenced to A phase with a scale factor of 30 millivolts per knot. Range of velocity inputs shall be between 70 and 600 knots, and shall be supplied by the Doppler/Air Mass Navigational Computer. These velocities will be used in the I-D and Slave modes. Input impedance shall be a minimum of 50,000 ohms.

3.4.8.4.1.3 North-South and East-West Velocity Corrections - These velocity corrections shall be supplied as 400 cycle AC signals referenced to A phase with a scale factor of 30 millivolts per knot. Range of velocity corrections shall be between zero and 25 knots, and shall be supplied by the AN/ASA-16 computer and will be summed with the N/S, E/W inertial velocities in the I-D mode.

3.4.8.4.1.4 AN/ASN-47 Doppler/Air Mass Navigation Computer Signal - This signal shall be supplied when the AN/ASN-47 Doppler Air Mass Navigation Computer is operating. The signal shall be +28 DC and shall be referenced to the ground supplied by the AN/ASN-47 Doppler/Air Mass Navigation Computer.

3.4.8.4.1.5 Heading Velocity - This velocity shall be supplied as a 400 cycle AC signal referenced to A phase with a scale factor of 30 millivolts per knot. Range of this velocity shall be as supplied by the AN/APN-122 or AN/APN-153 Doppler Radar. This velocity will be used in the I-D and Slave modes. The minimum load impedance shall be 25,000 ohms.

3.4.8.4.1.6 Drift Velocity - This velocity shall be supplied as a 400 cycle AC signal referenced to A phase with a scale factor of 30 millivolts per knot. Range of this velocity shall be between zero and 200 knots and shall be supplied by the AN/APN-122 of AN/APN-153 Doppler Radar. This velocity will be used in the I-D and Slave modes. The minimum load impedance shall be 25,000 ohms.

3.4.8.4.1.7 Doppler Memory Signal - A 28V DC memory signal referenced to the Doppler Radar ground shall be supplied from the AN/APN-122 or AN/APN-153 Doppler Radar when the Doppler Radar is in the Search mode. This signal shall be used in the I-D and Slave mode.

3.4.8.4.1.8 Synchro Excitation 115V-400 CPS - Synchro excitation voltage shall be provided from aircraft power.

3.4.8.4.1.9 Synchro Excitation 26V-400 CPS - Synchro excitation voltage shall be provided from aircraft power.

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3.4.8.4.1.10 Doppler Decoupling Signal - This signal will be supplied as a DC ground and shall be interrupted when the magnitude of the difference between the Inertially derived N-S or E-W velocity component and the corresponding Doppler derived N-S or E-W velocity component exceeds 15 feet per second ±2 ft/sec for more than 10 ±2 seconds and will remain on until the magnitude of this difference falls below 15 feet per second. (± 2ft/sec) for more than 10 ±2 seconds. This signal will be used in the I-D mode.

3.4.8.4.1.11 Input Voltages - The following voltages will be supplied by the power supply and relay unit and shall not be subject to interruptions:

45.0 ±1% Volts DC
20.0 ±1% Volts DC
28.0 ±1% Volts DC
-45.0 ±1% Volts DC

3.4.8.4.2 Outputs to Associated Equipment

3.4.8.4.2.1 Velocity Heading -

(1) Type 400 cps AC voltage
(2) Range - as supplied by AN/APN-122 or AN/APN-153
(3) Scale - 30 millivolts/knot
(4) Destination - D/AM Nav Computer
(5) Supplied in all modes when the AN/APN-122 or AN/APN-153 is on
(6) Load impedance: 50,000 ohms min.

3.4.8.4.2.2 Velocity, Drift

(1) Type 400 cps AC voltage
(2) Range - ±200 knots
(3) Scale - 30 millivolts/knot
(4) Destination - D/AM Nav Computer
(5) Supplied in all modes when the AN/APN-122 or AN/APN-153 is on
(6) Load impedance 50,000 ohms min.

3.4.8.4.2.3 Miles Travelled, North/South

(1) Type -1 each Case Size 11 or larger Synchro Transmitter; similar to Kearfott RS911-6G
(2) Destination AN/ASA-16
(3) Supplied only in I-D mode

3.4.8.4.2.4 Miles Travelled, East/West

(1) Type -1 each Case Size 11 or Larger Synchro Transmitter; similar to Kearfott RS911-6G
(2) Destination - AN/ASA-16
(3) Supplied only in I-D mode

3.4.8.4.2.5 True Heading

(1) 4 Power Synchro H_T Platform No. 2

(a) Type -1 each case size 8 or larger Synchro Transmitter; similar to Bendix AY-2015-384
(b) Output destination - D/AM Nav Comp
(c) Supplied in I-D, and Slave*

*Correct only if magnetic variation is properly set.

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(2) 20 Power Synchro (H_T 20 Power Repeater)

(a) Type - 1 each 20-power synchro transmitter; similar to Bendix AY-401-2
(b) Destination - ASQ maneuver monitor, MW-2, 4 Bearing Distance - Heading Indicators, ASA-16
(c) Supplied in I-D, slave *, and free (true heading channel supplies grid heading in free mode)

*Correct only if magnetic variation is properly set.

(3) 4 Power Synchro (H_T 4 Power Repeater)

(a) Type - I size 8, Litton Type 900141-68 or equivalent 26 V input, 11.8 output
(b) Destination - PT434 and PT396
(c) Supplied in I-D and slave*

*Correct only if magnetic variation is properly set

3.4.8.4.2.6 Magnetic Heading

(1) Type - 2 each Case Size 11 or larger Synchro Transmitters; similar to Kearfott 5RS-911-1H
(2) Destination - Two Horizontal Situation Indicators, AN/ARN-21A, or AN/ARN-52, APS-80, VORs I & II
(3) Supplied in I-D, slave and compass modes.

3.4.8.4.2.7 Pitch Angle

(1) Type - 2 each size 25 gimbal transmitter synchros, Litton 900141-75 or equivalent.
(2) Destination

1. Synchro No. 1 - AN/APN-122(V) or AN/APN-153(V) MM-4 and MX-2230
2. Synchro No. 2 - APS-80, 2 each, Exc 115 volts

(3) Supplied in all modes.

3.4.8.4.2.8 Roll Angle

(1) Type - 2 each size 25 gimbal transmitter synchros; Litton 900141-75 or equivalent
(2) Destination

1. Synchro No. 1 - AN/APN-122(V) or AN/APN-153(V), MM-4, and MX 2230.
2. Synchro No. 2 - APS-80, 2 each, Exc 115 volts

(3) Supplied in all modes

3.4.8.4.2.9 Clutched Magnetic - Heading

(1) Type -1 each Clutched Synchro, similar to Bendix CP21-A-1
(2) Excitation - Provided by autopilot, 26 volt, 400 CPS, phase VNA-30 degrees to synchro and 28V DC to clutch
(3) Supplied in all modes except the synchro clutch is disengaged when slewing in the free mode.

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3.4.8.4.2.10 Memory Signal - This equipment shall provide a memory signal output to the D/AM Nav Comp. whenever the Doppler output is not present.

3.4.8.4.2.11 Magnetic Variation

(1) Type - 1 size 8, synchro transmitter; similar to Kearfott 5R1011
(2) Excitation -----26 VAC supplied from fwd junction box
(3) Supplied in all modes.

3.4.8.4.2.12 North-South and East-West Velocities

(1) 400 cycle AC 30 mv/knot, reference to A phase
(2) to ASA-16 and Ground Speed Drift Angle Indicator
(3) Supplied in I-D mode
(4) Load impedance 50,000 ohms min.

3.4.8.4.2.13 Rate of Turn Signal

(1) Supplied as set of relay contacts with 3 terminals, normally open common and normally closed, changing when R/T exceeds 18 ± 3 degrees/min.
(2) In I/D, slave and free modes
(3) Supplied to D/AM Nav Computer

3.4.8.4.2.14 Inertial - Doppler Mode Signal

(1) 28 V DC when the system is in the I-D mode.

3.4.8.4.2.15 Free Mode Signal

(1) 28 V DC when the system is in the Free Mode.

3.4.8.4.2.16 Flux Valve Excitation - Flux valve excitation shall be provided for the Remote Magnetic Compass Transmitter 23.5V ±1, 400 cps.

3.4.8.4.2.17 Inertial System Failure Indication - An open circuit indicates failure.

3.4.8.4.2.18 Magnitude of the Difference between Doppler Derived North-South Velocities and Inertially Derived North-South Velocities - This output shall be provided as a D. C. potential with a voltage scale factor of 10 millivolts per foot per second when differences exist between the Inertially Derived North-South and Doppler Derived North-South Velocities.

3.4.8.4.2.19 Magnitude of the Difference between Doppler Derived East-West Velocities and Inertially Derived East-West Velocities - This output shall be provided as a D. C. potential with a voltage scale factor of 10 millivolts per foot per second when differences exist between the Inertially Derived East-West and Doppler Derived East-West Velocities.

3.4.9 Mounting MT-2961/ASN-42 - This unit shall meet the following requirements.

3.4.9.1 Function - The signal data converter mount serves to isolate the signal data converter from the aircraft vibrations.

3.4.9.2 Form Factor - The dimensions of the signal data connector mount shall not exceed 1.15 inches high, 9.07 inches wide and 16.19 inches long.

3.4.9.3 Weight - The weight of the signal data converter mount shall not exceed 3.3 pounds.

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3.4.10 Amplifier, Electronic Control AM-3113/ASN-42 - This unit shall meet the following requirements:

3.4.10.1 Function - The Electronic Control Amplifier contains the electronic amplifiers required for the stabilization of the inertial platform and for nulling of accelerometers. It also includes the 5 kc power supply, gyro spin motor power supply, gyro temperature controller, and circuits used in initial alignment and inertial system failure detection.

3.4.10.2 Form Factor - The dimensions of the electronic control amplifier shall not exceed 7.00 inches high, 6.50 inches wide and 14.13 inches long.

3.4.10.3 Weight - The weight of the electronic control amplifier shall not exceed 21.2 pounds.

3.4.10.4 Electrical Connections - The following voltages are required from the power supply and relay unit and shall not be subject to interruptions:

45.0 ±1% Volts DC
20.0 ±1% Volts DC
28.0 ±1% Volts DC
-45.0 ±1% Volts DC

3.4.11 Mounting MT-2960/ASN-42 - This unit shall meet the following requirements:

3.4.11.1 Function - The electronic control amplifier mount isolates the electronic control amplifier from aircraft vibration.

3.4.11.2 Form Factor- The dimensions of the electronic control amplifier mount shall not exceed 1.38 inches high, 7.25 inches wide and 14.00 inches long.

3.4.11.3 Weight - The weight of the electronic control amplifier mount shall not exceed 3.3 pounds.

3.4.12 Control, Gyroscope Assembly C-3832/ASN-42 - This unit shall meet the following requirements:

3.4.12.1 Function - This unit contains the following controls and displays: The latitude indicator, longitude indicator, latitude set control and longitude set control. These controls and displays permit the operator to set and readout aircraft present latitude and present longitude.

3.4.12.2 Form Factor - The dimensions shall not exceed 6.00 inches high, 5.75 inches wide and 6.60 inches long.

3.4.12.3 Weight - The weight shall not exceed 8.0 pounds.

3.4.12.4 Controls and Indicators - The Gyroscope Assembly Control shall include the following controls and displays:

3.4.12.4.1 Latitude Indicator - This display permits readout of present latitude.

3.4.12.4.2 Longitude Indicator - This display permits readout of present longitude.

3.4.12.4.3 Latitude Set Control - This control permits setting of present latitude.

3.4.12.4.4 Longitude Set Control - This control permits setting of present longitude.

3.4.13 Control, Inertial Navigation C-4061/ASN-42 - This unit shall meet the following requirements:

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3.4.13.1 Function - The Inertial Navigation Controller contains the following controls and displays: The mode control, slew control, latitude set control, sync control, sync indicator, and align indicator. These controls and displays permit the operator to properly select the various operational modes and to operate the system in these modes.

3.4.13.2 Form Factor - The dimensions of the inertial navigation control shall not exceed 3.00 inches high, 5.75 inches wide and 3.00 inches long.

3.4.13.3 Weight - The weight of the inertial navigation control shall not exceed 2.0 pounds.

3.4.13.4 Controls and Indicators - The Inertial Navigation Control snall include the following controls and displays.

3.4.13.4.1 Mode Control - This control selects the operational modes of the system.

3.4.13.4.2 Slew Control - This control permits slewing of the free mode heading output.

3.4.13.4.3 Latitude Set - This control permits setting of present latitude.

3.4.13.4.4 Sync - This control permits rapid adjustment of the magnetic heading output in the slave mode.

3.4.13.4.5 Sync Indicator - This display indicates that flux valve heading is being properly followed in the slave mode.

3.4.13.4.6 Align Indicator - This display indicates that the system is being aligned.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered at the contractors expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

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4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractors expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitiative results for all acceptance tests.

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Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 5 to 100 cps (excluding resonant points) and a minimum amplitude of ±3 g's except that the gyroscope assembly shall be vibrated at a maximum of ±2 g's

The equipment shall be vibrated (without vibration isolators except that the gyroscope assembly shall be vibrated on its vibration isolator) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure II from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be III from Table I of Specification MIL-R-23094 with the following exceptions:

chamber temperature	-35°C to +35°C
vibration	±2G at any non resonant frequency between 5 and 100 cps measured at the mounting points of the equipment except that the gyroscope assembly shall be measured at the mounting points of the vibration mount.

During each cycle the equipment shall be operated during the entire heating period.

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4.4.3.3 Accept-Reject Criteria - Figure 14 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Qualification Plan. Figure 16 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Sampling Plan.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 4 hours during each cycle.

4.4.3.5 Performance Characteristics to be Measured -

Characteristic	Spec Requirement	Acceptance Requirement (if lower)	Time in cycle that measurement is to be made
latitude and longitude	2 nautical miles per hour CEP	Not to exceed 5 nautical miles per hour.	Measurement is to be made for 2 hours after completion of alignment in the first cycle and each 5th cycle thereafter.

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.7 Preventive Maintenance - During the period of the tests the following preventive maintenance measures may be performed upon the equipment:

Measure	Time
Reset gyro and accelerometer biases.	Each 25 hours

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ±5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contractor order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The AN/ASN-42 Navigational Computer Set shall be integrated with a Doppler System to achieve Doppler damping and must operate in any of four modes: Inertial Doppler, Slave, Free, and Compass. In the Inertial-Doppler mode the system must continually provide aircraft heading and drift velocities, magnetic and true heading, magnetic variation, present longitude and latitude, north-south and east-west velocities, north-south and east-west miles travelled, and aircraft pitch and roll during extended flights over land and ocean. In the slave mode, the system must provide magnetic heading, and aircraft pitch and roll. If variation is set into the system, it must provide true heading. In the free mode, the system must provide grid heading, and aircraft pitch and roll. In the compass mode, the system must provide unstabilized magnetic heading from the flux valve and aircraft pitch and roll, except that if the inertial system fails, only unstabilized magnetic heading will be provided by the inertial system.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 V DC and 115 ±1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimun. size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, may be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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6.7 An asterisk "*" preceding a paragraph number denotes a change made in that paragraph from the previous version of the document. (See paragraph 4 of the Explanation page i.)

6.8 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

NOTE: Additional paragraphs may be included for Definitions or other pertinent information.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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